CHAPTER 2

CPU Performance

CPU time · CPI · clock rate · MIPS · weighted CPI · Amdahl's law · speedup

QUICK TRICK

CPU time = IC × CPI × T_{clk} = IC × CPI / f. Weighted CPI = Σ (fraction$_i$ × CPI$_i$). **Speedup = old time ÷ new time**. Amdahl: **S = 1 / [(1 − f) + f / k]**.

2.1 What does "faster" mean?

● **LEVEL 1 · BEGINNER**

The problem. Which computer is better? **The idea.** Compare the **time** to run the same program. **Analogy.** Two cars on one road: time taken is what counts, not how loud the engine is.

- **Latency (response time):** time for one task. **Throughput:** tasks per unit time.
- **Clock cycle time** T_{clk} = 1/f. At 2 GHz, T_{clk} = 0.5 ns.

2.2 The CPU time equation

● **LEVEL 2 · EXAM**

```
CPU time = Instruction count (IC) x Cycles per instruction (CPI) x Clock cycle time
         = IC x CPI / f
Total cycles = IC x CPI          MIPS = f / (CPI x 10^6) = IC / (CPU time x 10^6)
```

Factor	Changed by
IC	Compiler, instruction set
CPI	Organisation (pipeline, cache), instruction mix
T_{clk} (or f)	Technology, organisation

Weighted CPI. With instruction classes of fractions F_i and CPI$_i$: CPI = Σ F_i × CPI$_i$. **Example.** 50% ALU (CPI 1), 30% load/store (CPI 2), 20% branch (CPI 3): CPI = 0.5 + 0.6 + 0.6 = **1.7**.

Example. IC = 10^6, CPI = 2.5, f = 500 MHz: time = 10^6 × 2.5 / (5×10^8) = **5 ms**.

2.3 Amdahl's law

● **LEVEL 2 · EXAM**

If a fraction **f** of the execution time is made **k times faster**, the rest stays the same:

```
New time = (1 - f) + f/k            Speedup = 1 / [ (1 - f) + f/k ]
Maximum speedup (k -> infinity) = 1 / (1 - f)
```

Example. f = 0.8, k = 4: speedup = 1/(0.2 + 0.2) = **2.5**. Even with infinite k the limit is 1/0.2 = 5. **Two parts improved:** 40% by 2× and 30% by 3×: new time = 0.3 + 0.2 + 0.1 = 0.6, speedup = **1.67**.

QUICK TRICK

Amdahl uses fractions of **time before the improvement**, not fractions of instructions.

2.4 MIPS and benchmark means

● **LEVEL 3 · GATE**

- **MIPS** ignores what each instruction does, so it cannot compare different instruction sets (a CISC instruction does more than a RISC one).
- Averages: **arithmetic mean** for execution times, **harmonic mean** for rates (MIPS, MFLOPS), **geometric mean** for normalised ratios.
- Comparing machines: speedup of A over B = T_B / T_A.

2.5 Common mistakes

COMMON MISTAKE

- Using the clock rate alone to compare machines (CPI and IC matter too).
- Forgetting the unit: MHz = 10^6, GHz = 10^9, ns = 10^{-9} s.
- Applying Amdahl with f as a fraction of the **new** time.
- Averaging CPI values without weighting by instruction fractions.

GATE QUICK CHECK — Chapter 2

```
T = IC*CPI/f      CPI = sum(Fi*CPIi)      MIPS = f/(CPI*10^6)
Speedup = Told/Tnew      Amdahl = 1/((1-f)+f/k)      max = 1/(1-f)
IC x 0.8 and CPI x 1.1  ->  time x 0.88
```

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. Clock cycle time of a 2 GHz processor.

Q2. A program has 10^6 instructions, CPI 2.5, clock 500 MHz. CPU time?

Q3. Mix: 50% ALU (CPI 1), 30% load/store (CPI 2), 20% branch (CPI 3). Average CPI?

Q4. Why is MIPS a poor way to compare computers?

● **LEVEL 2 · EXAM**

Q5. 80% of a program is sped up 4×. Overall speedup? Maximum possible if that part became infinitely fast?

Q6. CPI = 1.7 on a 1 GHz machine. MIPS rating?

Q7. Machine A: CPI 2 at 1 GHz. Machine B: CPI 1.5 at 700 MHz. Same program. Which is faster and by how much?

Q8. A compiler change cuts IC by 20% but raises CPI by 10% (same clock). Speedup?

● **LEVEL 3 · GATE**

Q9. A component taking 40% of the time is made 2× faster and another taking 30% is made 3× faster. Overall speedup?

Q10. Program of 2×10^9 instructions runs in 4 s on a 2 GHz machine. Find CPI and MIPS.

Q11. Floating-point operations are 25% of the time. A new FP unit makes them 5× faster. Overall speedup?

Answers

Q	Solution
1	$1/(2\times10^9)$ = **0.5 ns**.
2	$10^6 \times 2.5 / 5\times10^8$ = **5 ms**.
3	0.5 + 0.6 + 0.6 = **1.7**.
4	Different instruction sets do different amounts of work per instruction.
5	1/(0.2 + 0.8/4) = **2.5**. Maximum = 1/0.2 = **5**.
6	1000/1.7 ≈ **588 MIPS**.
7	Per instruction: A = 2 ns, B = 1.5/0.7 = 2.143 ns. **A is faster** by 2.143/2 ≈ **1.07×**.
8	New time = 0.8 × 1.1 = 0.88 of old. Speedup = 1/0.88 ≈ **1.14**.
9	New time = 0.4/2 + 0.3/3 + 0.3 = 0.2 + 0.1 + 0.3 = 0.6. Speedup = **1.67**.
10	Cycles = $4 \times 2\times10^9 = 8\times10^9$; CPI = $8\times10^9/2\times10^9$ = **4**. MIPS = $2\times10^9/(4 \times 10^6)$ = **500**.
11	1/(0.75 + 0.25/5) = 1/0.8 = **1.25**.

Next: Chapter 3 — Number Representation and Arithmetic (2's complement, IEEE 754).

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